

EOD Report – Task 5: Parameterized Jenkins Job

1. Task Objective

Create a **parameterized Jenkins Pipeline job** that accepts user input before starting the build.

The parameters used were:

- ENVIRONMENT – Select deployment environment
- APP_VERSION – Enter application version

The pipeline contains three basic stages:

1. Build
2. Test
3. Deploy

For this task, the stages only display messages using echo.

2. Source Code

GitHub Repository:

https://github.com/Zeeshancloud15/Tomcat_app01.git

3. Jenkins Job Creation

Created a Jenkins Pipeline job named:

parameter-job

Configuration

Enabled:

This project is parameterized

Parameter 1 – Choice Parameter

Name:

ENVIRONMENT

Values:

dev

test

prod

Description:

Select deployment environment

Parameter 2 – String Parameter

Name:

APP_VERSION

Default value:

1.0

Description:

Enter application version

4. Pipeline Script

The basic parameterized Jenkins Pipeline used was:

```
pipeline {
    agent any

    parameters {
        choice(
            name: 'ENVIRONMENT',
            choices: ['dev', 'test', 'prod'],
            description: 'Select environment'
        )

        string(
            name: 'APP_VERSION',
            defaultValue: '1.0',
            description: 'Enter version'
        )
    }

    stages {
        stage('Build') {
            steps {
                echo "Build - ${params.APP_VERSION}"
            }
        }

        stage('Test') {
            steps {
                echo "Test - ${params.ENVIRONMENT}"
            }
        }

        stage('Deploy') {
            steps {
                echo "Deploy - ${params.ENVIRONMENT}"
            }
        }
    }
}
```

5. How Parameters Are Used

Jenkins parameters can be accessed using:
`params.ENVIRONMENT`

and:
`params.APP_VERSION`

Example:
`echo "Build - ${params.APP_VERSION}"`

If the user selects:
`ENVIRONMENT = dev`
`APP_VERSION = 1.0`

Jenkins displays:
`Build - 1.0`
`Test - dev`
`Deploy - dev`

6. Stages

Build

The Build stage displays the application version.

```
stage('Build') {  
    steps {  
        echo "Build - ${params.APP_VERSION}"  
    }  
}
```

Test

The Test stage displays the selected environment.

```
stage('Test') {  
    steps {  
        echo "Test - ${params.ENVIRONMENT}"  
    }  
}
```

Deploy

The Deploy stage displays the selected environment.

```
stage('Deploy') {  
    steps {  
        echo "Deploy - ${params.ENVIRONMENT}"  
    }  
}
```

```
}  
}
```

7. Issue Faced During Testing

Initially, Maven was used in the pipeline for building the application.

The Jenkins console showed:

```
mvn: command not found
```

Resolution

The Jenkins Maven installation was configured as:

```
maven-integration
```

The Maven tool was then configured in the pipeline.

Later, the old application produced:

```
Source option 6 is no longer supported.
```

```
Target option 6 is no longer supported.
```

This was identified as an **old Java compiler configuration in the application**, not a Jenkins parameter problem.

For the parameterized-job task, the pipeline was simplified to use only echo messages for Build, Test, and Deploy.

8. Final Result

The parameterized Jenkins job was successfully configured with:

```
ENVIRONMENT  
├── dev  
├── test  
└── prod
```

```
APP_VERSION  
└── 1.0
```

The pipeline successfully demonstrates:

Parameters

↓

Build

↓

Test

↓

Deploy

The task demonstrates how Jenkins parameters can be created and accessed inside a Declarative Pipeline.

9. Task Status

Jenkins / parameter-job

Status

Changes

Build with Parameters

Delete Pipeline

Configure

Rename

Full Stage View

Stages

Pipeline Syntax

parameter-job

Add description

Stage View

Average stage times:
(full run time: ~477ms)

	Build	Test	Deploy
#10 Oct 06 17:52 No Changes	61ms	45ms	46ms

Permalinks

- Last build (#10), 0.26 sec ago
- Last stable build (#9), 1 min 43 sec ago
- Last successful build (#9), 1 min 43 sec ago
- Last failed build (#7), 5 min 5 sec ago
- Last unsuccessful build (#7), 5 min 5 sec ago
- Last completed build (#9), 1 min 43 sec ago

Builds >

Filter

Today

#10 12:22 PM

Choice Parameter ?

Name ?

ENVIRONMENT

Choices ?

dev
test
prod

Description ?

Select environment

String Parameter ?

Name ?

APP_VERSION

Default Value ?

1.0

Description ?

Enter version

Plain text Preview